Sustainable Life Management

ependent Resource Allocation Under Uncertainty

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Agenda

Background

Motivation

- Cognitive overload and limited attention make time allocation fragile [Robinson, 2024, Grant, 2019].
- Sustainable planning needs explicit tradeoffs under finite time [Hart, 2024].
- Team settings add coordination and capacity coupling across agents.

Problem Formulation

Problem Setting

- Tasks are items, agents are knapsacks with capacities C_j.
- Task to agent assignments face stochastic sizes S_{ij} and returns R_{ij}.
- Overload is controlled by chance constraints on capacity violations.

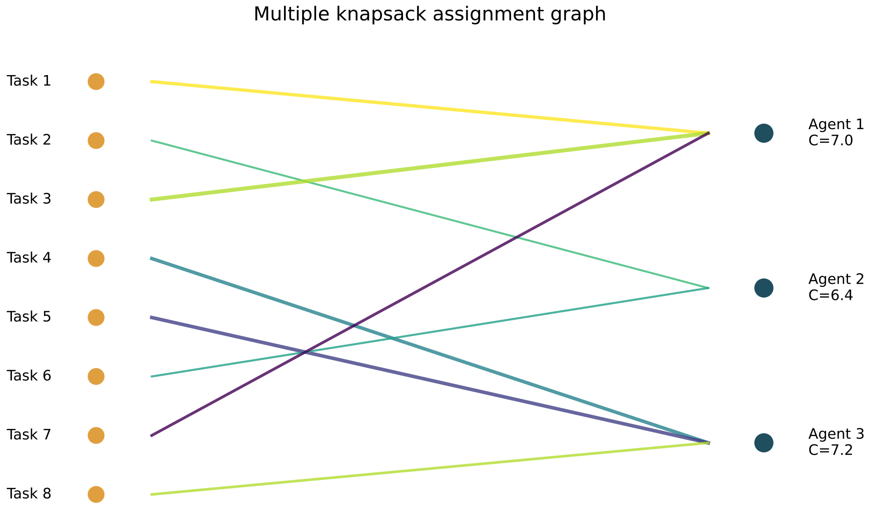
Stochastic Multiple Knapsack

$$
\begin{aligned}
\max \quad & \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} \mathbb{E}\left[R_{ij}\right] x_{ij} \\
\text{s.t.} \quad & \sum_{j \in \mathcal{J}} x_{ij} \leq 1 \quad \forall i \in \mathcal{I} \\
& \mathbb{P}\left(\sum_{i \in \mathcal{I}} S_{ij} x_{ij} \leq C_j \right) \geq 1 - \epsilon_j \quad \forall j \in \mathcal{J}.
\end{aligned}
$$

Sizes and returns can be correlated, and risk is limited through chance constraints.

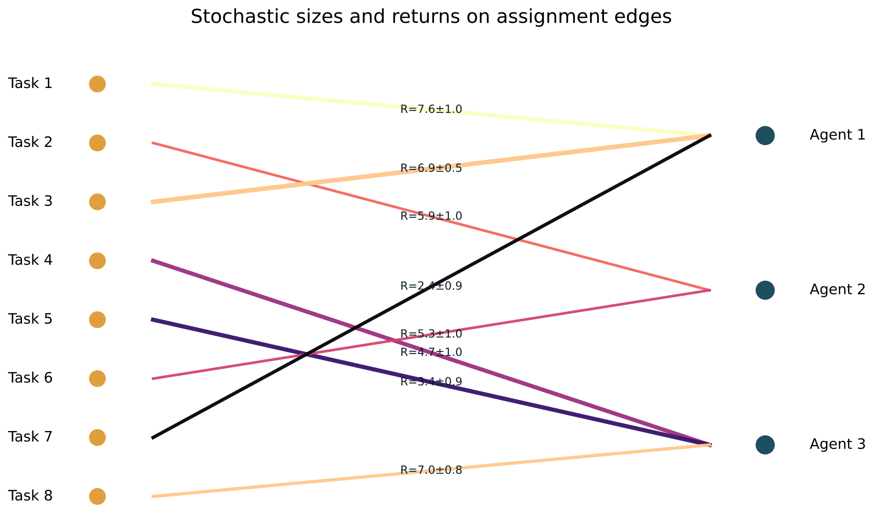
Visual Intuition

Multiple Knapsack Structure



Agents

Stochastic Sizes and Returns



Each

Methodological Context

Related Methods

- Scenario based cuts for stochastic knapsack [Büyüktahtakın, 2023].
- Distributionally robust knapsack formulations [Bos et al., 2024, Ding et al., 2022].
- Sample average approximation for two stage programs [Chen and Luedtke, 2022].
- Learning accelerated L shaped heuristics [Larsen et al., 2024].
- Stochastic assignment in project settings [Rezaeian et al., 2024].

Takeaways

Key Takeaways

- The formulation unifies cognitive resource allocation with stochastic knapsack structure.
- Chance constraints control overload risk while retaining assignment flexibility.
- Decomposition and data driven methods are a natural fit for scaling the model.

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